

Stock Market Analysis System with ML-Driven Recommendations

Project Proposal for Software Engineering Lab

Submitted to: [Instructor Name] • [Institution Name]

[Team Member 1], CSED* [Team Member 2], CSED† [Team Member 3], CSED‡

August 20, 2026

*Project Manager, Roll No: [xxxxxxx], Email: [name1]_be24@thapar.edu †Frontend Developer, Roll No: [xxxxxxx], Email: [name2]_be24@thapar.edu ‡Backend Developer, Roll No: [xxxxxxx], Email: [name3]_be24@thapar.edu

1 Higher-order goal ...secondary framing

This project aims to improve retail-investor decision-making by consolidating research, trend analysis, and practice-trading into a single academic tool. While the broader goal is financial literacy, the immediate engineering objective is an ML-backed stock analysis and recommendation system for the Indian stock market.

2 Time-to-value ...primary heuristic

- Deliver a working core (auth, search, charting) before layering ML-driven recommendations.
- Use weekly Agile Scrum sprints to get fast feedback and keep release friction low.
- Treat ML-based prediction as an enhanced capability, introduced once the base pipeline is stable.

3 Problem Statement ...

Retail investors and finance students rely on disconnected tools — brokerage apps, news sites, spreadsheets — to research a stock before deciding. Two gaps stand out:

- **Fragmentation:** no single, lightweight tool combines live/historical price data, trend visualization, and a safe practice-trading environment.
- **Interpretability:** raw price data and indicators are hard to translate into a decision without factoring in an individual's risk appetite, horizon, and capital.

4 Proposed Solution ...software proposition

4.1 Overview. A web-based platform with:

- Unified dashboard — search, compare, and track stocks with market indices.
- Historical trend charts (line and candlestick) with technical indicators.
- ML-driven pipeline that scores a stock's outlook as Favorable / Neutral / Unfavorable, personalized by the user's risk preference, investment horizon, and capital.
- Paper-trading simulation — virtual capital, simulated Buy/Sell/Hold, live portfolio value and profit/loss tracking.
- Watchlists and price alerts for ongoing monitoring.

4.2 Core workflow.

- User selects or searches for an Indian stock.
- Backend retrieves historical and market data (cached to respect API rate limits).
- Feature engineering generates technical/market features for the ML model.
- The model estimates outlook; this is combined with the user's risk context to produce a personalized recommendation with supporting insight.
- The user can act on it via simulated Buy/Sell/Hold, and the portfolio, P&L, and performance are tracked for evaluation.

5 Solution Approach ...engineering focus, with foresight

A standard, stateless 3-tier web architecture keeps the system correct and scalable without novel ML research:

- **Frontend:** React.js with a charting library (Chart.js / Recharts) and Tailwind CSS.
- **Backend:** Node.js/Express (or Django) REST API — auth (JWT), caching, and the feature-engineering / ML-prediction pipeline; stateless design for horizontal scaling.

- **Data:** PostgreSQL/MongoDB for users, portfolios, and transactions; Alpha Vantage / Yahoo Finance for market data, cached to stay within free-tier rate limits.
- **Deployment:** managed, free-tier-friendly hosting (Render/Railway/Vercel) with CI via GitHub Actions.

6 Evaluation Criterion ...*measurable and attributable*

Primary: quote/recommendation response time — target median ≤ 2 seconds under normal load, measured at the API layer.

Secondary: demo-period availability $\geq 95\%$; simulated portfolio P&L accuracy against recorded transactions; user-reported clarity of the recommendation (1–5 scale).

7 Project Scope and Deliverables ...*iteration-friendly*

Initial deliverable (Sprint 1): project scaffolding, low-fidelity wireframes, database schema draft, JWT auth flow, and a proof-of-concept call to the external stock API.

Subsequent deliverables: watchlist and historical charting (core); ML prediction pipeline and paper-trading simulation (enhanced); price alerts, stock comparison, and news feed (enhanced/stretch).

8 Risks and Mitigations ...

- **API rate limits** block dev/demo — caching layer plus a mock-data fallback.
- **Scope creep** — freeze scope to core features after Sprint 3.
- **ML complexity underestimated** — start with a simple baseline (moving-average / linear regression) before more advanced models.
- **Uneven contribution** — rotate roles each sprint; require PR review before merge.

9 Summary ...

This proposal describes a deployable, ML-centric stock analysis platform for the Indian market that pairs a personalized recommendation pipeline with a risk-free paper-trading simulator. It emphasizes time-to-value through weekly iterations and measurable evaluation criteria, prioritizing a stable core before layering ML-driven and enhanced features.